Midmark Multiparameter Monitor
Clinical In-Service Video

Visuals: Live video, video/presentation/etc. Like this: https://vimeo.com/568997602/8b66771f23
Target Date: Photo/video capture Monday, May 20; testimonial capture Monday, July 22; target completion (est. December 2024)
Goal: Provide a shareable resource to reduce the quantity of live in-services, while providing the customer the same value the clinical team could offer live.

Blue – Professional voiceover; Footage - slide with on-screen text
Orange – Professional voiceover; Footage - still Image with on-screen animation highlighting voiceover points.
Green – Professional voiceover; Footage – still images and live video, shoot both options for editing flexibility
Purple – Mandi testimonial live footage

	VOICEOVER	VISUAL NEEDS ***Bold is on-screen text**
Intro :05	No voiceover	Midmark video intro animation: 
Intro	Welcome to the Midmark Multiparameter Monitor clinical in-service video. You will learn how to navigate the status bar, quick access buttons, and parameter settings. Status	Title Slide: ***Midmark Multiparameter Monitors*** ***Clinical In-Service Video*** Show images of both monitors
Overview	Upon startup, the monitor's user interface will display the status bar, parameter channels, and 4 or 5 quick access buttons depending on monitor model.	***Overview*** Still images + navigation ***Note:*** ***12" Monitors – 5 Quick Access Buttons***

		8" Monitors – 4 Quick Access Buttons
Status Bar	Let's start with the status bar, moving from left to right. The alarm message center displays technical and physiological alarms. The color of the alarm indicator light, located at the top right of the monitor, indicates alarm severity. To display the alarm message center, select an alarm. Here a parameter alarm can be paused for 120 seconds or reset to silence until the monitor is restarted. Cycle through all alarms by selecting the next alarm icon. For technical alarms, select the alarm label to open the troubleshooting guide. Next is the patient setup menu. Select the patient icon to select species and enter patient info. The patient ID is required to use the Midmark Anesthetic Record Interface. The patient setup menu also allows patient alarms to be cleared, the ability to view tabular and NIBP trends, and to discharge the patient at the end of a procedure. Once the patient information is entered, select okay to save the information. The status bar icon will display patient name and species. Next is the volume setup menu. Select the volume icon to adjust volume ranges for alarm, HR beat, pulse, and "touch" from 0-5. When alarm volume is set to 0, the icon will change to indicate an alarm has been silenced. For safety, alarm volume will reset upon monitor restart. Next is the Time/Date setup. Select the date or time options to adjust accordingly. Next is the battery icon with current battery status. Last on the status bar is the network connection. This will display the LAN or Wi-Fi connection status when subscribed to the Midmark Anesthetic Record Interface.	***Alarm Message Center*** ***Alarm Severities*** ***Blue = Low*** ***Yellow = Medium*** ***Red = High*** Show still image of troubleshooting guide example ***Patient Setup*** ***Note: Patient ID is required to leverage the Midmark Anesthetic Record Interface. midmark.com/ari*** ***Volume Setup*** ***Time/Date*** ***Battery*** ***Network Connection***

Quick Access Buttons	The monitor has 4 or 5 quick-access buttons, wave the model purchased. Moving from left to right, first is the alarm reset option to silence technical and physiological alarms. Next is the button to pause alarms for 120 seconds. The NIBP start icon will start or stop a blood pressure measurement. The print icon will start or stop printing on 12-inch models with integrated printers. The last icon is the menu button which will open the on-screen menu options.	***Note:*** ***12" Monitors – 5 Quick Access Buttons*** ***8" Monitors – 4 Quick Access Buttons***
Menu Dock	Moving from left to right, first is the patient setup menu as shown in the status bar. Next is the suspend option. This can be selected on or off. This function suspends monitoring and alarms while setting up for a procedure, but the patient is not ready for monitoring. It will reduce excess data recordings. The freeze icon freezes the waveforms across the screen for further analysis. Tap the icon to freeze or unfreeze. The screen lock icon will lock the touch screen functionality. Tap the icon to lock or unlock. The trends icon will open views to tabular, graphic, or NIBP trends. Tabular trend displays a table of historical values. Graphic trend displays historical data in a customized trend graphic. NIBP trend displays historical NIBP data. The displays icon allows the selection of different display modes. The monitor defaults to 3 channels with tabular trend option. Press the "info" icon to preview the pre-set	***Patient Setup Menu*** ***Suspend*** ***Freeze*** ***Screen Lock*** ***Trends*** ***Displays***

	display options and select “use this display” to prompt the change. 7 Channel ECG is available when monitoring 5 lead ECG and 5-8 Channel is available when monitoring IBP and Multigas. To create a customized display, select the custom icon at the bottom left corner and choose “Create User”. Select species and a New User profile will appear. Press Edit to enter the Name, configure the alarm limits, and to customize your display. You can create up to 7 custom users. Next is the Print icon. A secondary option to start or stop printing. In the Settings Menu, navigate to the maintenance menu, view recently recorded waveforms in wave recall, export data to a USB drive, initiate demo mode, access module or printer setup menus, adjust screen brightness or adjust date and time.	***Custom Displays*** ***Print*** ***Settings Menu***
Parameter Settings	The individual parameters can be accessed in the settings menu or by selecting the parameter on the screen. Starting with ECG. Select the ECG waveform to open the ECG setup menu. The monitor uses ECG leads, or electrodes to produce a waveform from the electrical conduction of the heart. The heart rate is then displayed based on the QRS waveforms produced. It’s important to remember that ECG is used to monitor heart rate and rhythm, not function. For the HR Source, choose between Auto, ECG, or SPO2 to hear the heart rate.	***ECG***

	ECG 1 is the chosen electrical pathway, or lead. Lead II is typically used for anesthetic monitoring and produces an upright P-QRST complex. While lead I and III are less commonly used for anesthesia, they can be adjusted based on patient circumstances. Wave gain increases or decreases the amplitude or amount of signal coming from the patient's heart. The signal volume impacts the waveform size and typically is related to the patient's heart size. The gain may be increased or decreased as needed to adjust the size of the waveform on the display. Wave speed is the speed of the waveform going across the screen in millimeters per second. The Cascade function allows the ECG waveform to continue onto a second line of display. If ECG Catalog is being displayed, Cascade will not be an option. The ECG catalog allows you to quickly scroll through 10 of the most common arrhythmias occurring in patients under anesthesia. The filter setting helps remove electrical interference. The surgery default is recommended for general anesthetic use where the other electrical devices are commonly used such as fluid pumps and patient warmers. The Diagnostic setting is unfiltered. The Monitor setting filters out low-level interference. Filter, Diagnostic and Monitor settings are commonly used for diagnostic ECG. High sensitivity allows for detection of weak or low signals. Lead Type is defaulted to 3- Lead. 5- lead is for optional 5-Lead wire sets. HR Beat Volume may be adjusted.	**Note: Too small of a waveform may result in an underestimated HR, or even a false asystole message. Too large of a waveform may cause double counting. If an asystole error message is occurring check your patient first.** **Note: Larger patients tend to have a smaller gain number and smaller patients tend to have a higher gain number.**

	The heart rate upper and lower limits and alarm priority can be adjusted in the alarms tab. The trend tab displays the ECG trend is graphic and numeric forms. For ECG accessories, it is recommended to use Redux gel or similar to ensure enough electrolyte material for a quality ECG. For cleaning, ECG leads and cables can be wiped down with alcohol or mild soap and water. ECG clips can be cleaned with alcohol. For storage, loosely coil the accessories to prevent breakage in the wiring. Next is SPO2. Select the SPO2 waveform to open the SPO2 setup menu. SpO2 monitoring estimates the arterial blood oxygen saturation and produces a PR with plethysmograph waveform. Adequate pulsatile blood flow, or perfusion, impacts these values. Poor perfusion, which may be due to hypotension, vasoconstriction, or hypothermia, can cause difficulty obtaining readings and decrease accuracy. Wave speed is the desired speed of the plethysmograph waveform across the screen. Wave mode allows to either fill in the graph or keep as a line. SatSeconds Limit is a secondary alarm when chosen to help filter out insignificant alarms. Pulse volume may also be adjusted in this menu. The SPO2 and pulse rate upper and lower limits and alarm priority can be adjusted in the alarms tab.	***ECG Accessories*** ***SpO2***

	The trend tab displays the SPO2 trend in graphic and numeric forms. The Spo2 lingual sensor includes both a large and small clip. Choose the appropriate size to ensure the sensor's optical components are parallel to one another. Deeply pigmented areas and artificial light directly on the sensor may impact readings. It is also recommended to move the sensor location every 10 minutes. For cleaning, SPO2 accessories may be wiped down with alcohol or mild soap and water. For storage, loosely coil the SPO2 cords to prevent breakage in the wiring. Temperature has 2 ports, either of which or both can be used. If using two temperature probes, the temperature delta will display the difference between the two values. For temperature, it is important to remember to monitor for trends. A temperature probe may be used in the esophagus or rectum. Esophageal temperatures are closer to core body temperature since the placement is over the heart and the core vessels of the body. However, avoid using a temperature probe in the esophagus during dental or oral procedures to reduce the risk of damage to the probe. Note that rectal temperatures may result in false high values due to fecal material insulating the probe. Select the temperature box to open the temperature setup menu. Units are defaulted to Fahrenheit with the option to change to Celsius.	**SPO2 Accessories** ***Note: The lingual SpO_2 sensor can be clipped to the patient's lip, toe, ear, prepuce or vulva.*** ***Note: Moving the sensor location every 10 minutes is recommended.*** ***Note: Avoid cleaning solutions that can be caustic like chlorhexidine.*** ***Temperature*** ***Note: Avoid using a temperature probe in the esophagus during oral procedures as probe damage can occur.***

	The upper and lower limits and alarm priority can be adjusted in the alarms tab. The trend tab displays the temperature trend in graphic and numeric forms. For cleaning, the temperature probe may be wiped down with mild detergent and water or 70% alcohol solution. For storage, loosely coil the temperature probe cords to prevent breakage in the wiring. The Non-invasive blood pressure uses oscillometric technology to obtain a systolic, diastolic and mean arterial pressure, also known as the MAP. Patient positioning and cuff sizing are the most common causes for inaccurate readings. The Blood Pressure Cuff Selector is used to determine the correct cuff size by wrapping it around the patient's limb. If in-between sizes, choose the next size up for accuracy. Cuffs should be placed over the artery on the metatarsus or metacarpus of the limb. The cuff should be on the same horizontal plane as the heart, avoiding limbs with catheters or surgical ties. In sternal or dorsal recumbency this is usually easy to achieve but if in lateral recumbency, one of the upside limbs should be chosen. If you position the cuff below heart level, the reading will be erroneously high. If you place the cuff above the heart, the reading will be low. This is true for both anesthetized and awake patients. Select the NIBP waveform to open the NIBP setup menu. Cuff size relates to the cuff option chosen. Small is for sizes 1-5, large is for size 8 or 10 cuffs.	***Temperature Accessories*** ***NIBP*** ***Note: Cuffs should be 30-40% of the patient's limb circumference.***

	Units are set in millimeters of mercury, with the option to change to kilopascals. Interval is defaulted to manual but may be changed to a set number of minutes for the monitor to automatically obtain measurements. The NIBP Best Practices Reminder may be turned on or off. This is a list of recommendations for proper blood pressure technique in both awake and anesthetized patients. Initial pressure is the inflation rate applied to the cuff when the initial measurement begins. The monitor and will increase incrementally, if needed, with each attempt at measurement until it is able to pressurize at a higher value than the patient's systolic pressure. If hypertension is suspected, increasing this setting will cut down on the time it takes to achieve a reading. Screen Mode will obtain 5 NIBP measurements, discard the outlier, then calculate the average values. Screen mode is activated in the menu or by holding the quick access icon. Start measurement will begin the first NIBP measurement. Select stop measurement to discontinue. Continuous measurement takes constant readings for 5 minutes with a 5 second delay in between readings. This is used for emergency situations vs standard trend monitoring. Select alarm tab to edit upper and lower limits, and alarm priority. The trend tab displays the NIBP trend in graphic and numeric forms.	**Note: This is useful for taking readings on awake patients and follows ACVIM guidelines.** ***Note: The ACVA states that readings should be taken at least once every 5 min.***

	NIBP cuffs are disposable. Avoid using tape or vet wrap to secure the cuff in place. This can prevent the bladder of the cuff from fully inflating and alter results. Next, we will discuss Co2 monitoring, or capnography. Capnography is a non-invasive means to evaluate patient ventilation. Monitoring capnography can help identify correct endotracheal tube placement, metabolic changes, poor cardiac output, ET tube plugs or kinks, exhausted soda lime, and anesthesia machine malfunctions within 2 breaths. Metabolic changes and decreases in cardiac output may be early warning signs of impending cardiac arrest. If a CO2 module was purchased, it must be activated in the menu. Press menu. Select settings. Select modules. Expand the drop-down option next to CO2 and select Masimo or Respironics. The monitor will display a capnogram, as well as three values: end-tidal CO2, inspiratory CO2 and respiratory rate. Apnea time is a secondary alarm that will alert no respirations detected after the set time. Unit is set to millimeters of mercury and may be changed to kilopascals or percentage. Wave speed is the speed of the waveform moving across the screen in millimeters per second. Wave mode changes the capnogram between fill or line. Fill will display the capnogram in more detail. Show allows you to toggle between showing the CO2 waveform or impedance respiration waveform.	***NIBP Accessories*** ***CO2***

	Wave grid may be turned on to view gridlines within the CO2 waveform. Weekly, the CO2 Soda Lime Absorbent Reminder will display providing best practice reminders on replacing CO2 absorbent. The CO2 catalog allows you to quickly scroll through 10 of the most common arrhythmias occurring in patients under anesthesia. A CO2 sensor includes either disposable airway adapters or filter lines intended for single-patient use. For cleaning, the sensor may be wiped down with a mild detergent solution. Thoracic impedance, referred to as RESP on the display, produces a RR and waveform from movement detected from the ECG lead wires on the chest or esophageal ECG probe. Apnea time is a secondary alarm that will alert no respirations detected after the set time. Wave gain is the amplitude of the signal coming from the leads, increase or decrease as needed. Wave speed is the speed of the waveform moving across the screen in millimeters per second. The wave mode allows to choose between fill and line. RESP lead is the chosen lead wires to detect respiration movement. The sensitivity option allows to increase or decrease motion sensitivity.	***Note: If a CO2 sleeping error message is received while using Masimo® CO2, change the operating mode from standby to measure to begin monitoring.*** ***Masimo is a registered trademark of Masimo Corporation.*** ***CO_2 Accessories*** ***Note: The larger adapter/filterlines are suggest for patients with a size 4.5 ET tube or larger and the smaller for size 4.0 and under ET tubes.*** ***Respiration*** ***Note: Best practice is to monitor CO2 for accurate respiratory rate.***

	Show will be greyed out when not using a CO2 sensor. For cleaning, RESP leads can be wiped down with 70% alcohol or mild soap and water. For storage, loosely coil the accessories to prevent breakage in the wiring.	***Respiration Accessories***
Closing	You're now ready to use your new multiparameter monitor. Thank you for choosing Midmark as your trusted partner in vital signs monitoring. For more detailed information about your monitor, refer to the Midmark Multiparameter Monitor User Guide. For other assistance, contact Midmark Clinical Support.	***Thank you for choosing Midmark as your trusted partner in vital signs monitoring. Please reach out to the Midmark Clinical Support Team if you have any questions.*** ***1.844.856.1232, press 2***
Closing	No voiceover	Traditional Midmark closing slides. Midmark.com/monitors

Key Words: veterinary monitors, Midmark multiparameter monitor, Cardell veterinary monitor, Midmark Cardell, monitoring anesthesia.

Title: Midmark Multiparameter Monitor Clinical In-Service Video

Short Description: Learn more about the new Midmark Multiparameter Monitors. For more information, visit: https://www.midmark.com/monitors